

Capture the Flag 2: Widgets

Goal: Turn the single-card application into a simple multi-section interface.

Note from the instructor before you start:

These activities involve using your own problem solving and researching skills. We are strictly to implement NOT TO USE AI to generate the code for you. You may only use AI for RESEARCHING the syntax or usage, but NEVER THE CODE ITSELF.

Consequence: Immediate 0 score in your grade for this activity.

(I know what AI-generated code looks like, don't even think about it :>)

🚩 Flag 0 – Create Your Feature Branch

Objective: Create a feature branch before modifying your application. Name the branch feature/widgets

Screenshot:

- Terminal showing feature/widgets

```
PS C:\Users\admin\Documents\FlutterProjects\my_first_flutter_app> git branch
* feature/widget
  feature/widgets
  master
PS C:\Users\admin\Documents\FlutterProjects\my_first_flutter_app> |
```

🚩 Flag 1 – Break It Apart

Objective: Separate your profile information into individual Text widgets.

If your card currently looks like this:

```
Text(
  'Chiara Marie C. Canque\n'
  'BSIT - 3\n'
  'My First Flutter Application\n'
  'August 11, 2026',
)
```

Your task: Separate each piece of information into its own Text widget.

You should have something conceptually like:

Text → Name

Text → Course & Section

Text → Application Title

Text → Date

Do not worry about making it pretty yet.

Screenshot:

- Your code showing the separate Text widgets
- Your application
- My first activity already separated these into its own text widget.

```
// Full Name
const Text(
  'Maverick Tajanlangit',
  maxLines: 1,
  style: TextStyle(
    color: Colors.white,
    fontFamily: 'serif',
    fontSize: 30,
    fontWeight: FontWeight.bold,
    fontStyle: FontStyle.italic,
  ), // TextStyle
), // Text

const SizedBox(height: 15),

// Course & Section
const Text(
  'BS Information Technology - 2nd Year',
  textAlign: TextAlign.center,
  style: TextStyle(
    color: Colors.white,
    fontFamily: 'serif',
    fontSize: 19,
  ), // TextStyle
),
```

```
// Application Title
const Text(
  'My First Flutter Application',
  textAlign: TextAlign.center,
  style: TextStyle(
    color: Colors.white,
    fontFamily: 'serif',
    fontSize: 22,
    fontWeight: FontWeight.w600,
  ), // TextStyle
), // Text

const SizedBox(height: 15),

// Date
const Text(
  'August 11, 2026',
  style: TextStyle(
    color: Colors.white,
    fontFamily: 'serif',
    fontSize: 18,
  ), // TextStyle
), // Text
],
```



🚩 Flag 2 – Stack the Information

Objective: Arrange your individual Text widgets vertically.

Research how to use:

Column()

Place your information inside the Column.

Your card should now have:

Name

Course & Section

Application Title

Date

Screenshot:

- The Column in your code
- The four separate pieces of information displayed vertically
- This was also implemented from my last activity...

```
body: Center(  
  child: Column(  
    mainAxisAlignment: MainA
```



🚩 Flag 3 – Add Another Element

Objective: Add another individual widget to your Column.

Add something that belongs to your profile.

Examples:

Hobby

It must be its **own widget**, not additional text inside another Text.

Screenshot:

- New widget in your Column
- Updated application

```
// Hobby
const Text(
  'Hobby: Dancing',
  style: TextStyle(
    color: Colors.white,
    fontFamily: 'serif',
    fontSize: 18,
  ), // TextStyle
), // Text
```



Flag 4 – Put Two Things Side-by-Side

Objective: Discover the Row widget.

Create a new section containing **two pieces of information beside each other**.

For example:

Age: 21 Birthdate: August 11, 2003

Research:

Row()

Place the Row inside your existing Column.

Your structure should now resemble:

Column

└─ Text

└─ Text

└─ Text

└─ Text

└─ Text

└─ Row

└─ Text

└─ Text

Screenshot:

- Your widget structure/code
- The two items displayed side-by-side

```
child: Row(  
  mainAxisAlignment: MainAxisAlignment.spaceBetween,  
  children: [  
    const Text(  
      'Age: 20',  
      style: TextStyle(  
        color: Colors.white,  
        fontFamily: 'serif',  
        fontSize: 18,  
      ), // TextStyle  
    ), // Text  
    const Text(  
      'Birthdate: May 23, 2005',  
      style: TextStyle(  
        color: Colors.white,  
        fontFamily: 'serif',  
        fontSize: 18,  
      ), // TextStyle  
    ), // Text  
  ],  
),
```



Flag 5 – Give It Some Space

Objective: Improve the spacing between your widgets.

Research:

SizedBox()

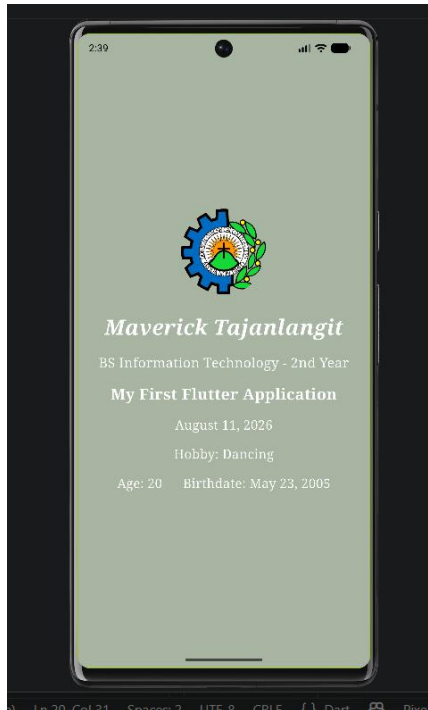
Add appropriate spacing between your:

- Image
- Name
- Course
- Other information
- Row

Challenge: Make the card easier to read **without changing its colors or fonts yet.**

Screenshot:

- SizedBox usage
- Updated application



```
const SizedBox(height: 25),
```

```
const SizedBox(height: 15),
```

```
const SizedBox(height: 15),
```

🚩 Flag 6 – Make It Look Better

Objective: Apply styling to your individual widgets.

Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

TextStyle()

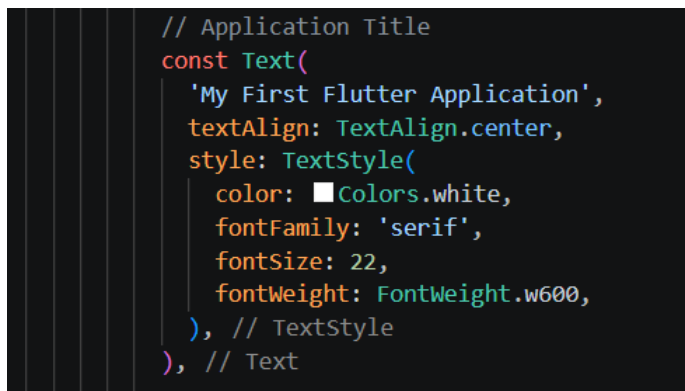
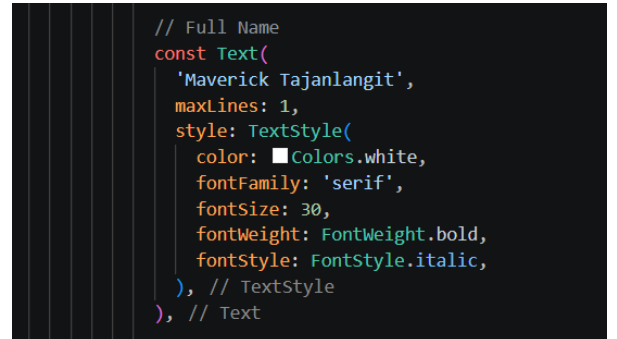
Apply different styling to at least **three** elements.

For example:

- Name → larger and bold
- Course → medium size
- Date → smaller
- Application title → italic

Screenshot:

- Your TextStyle changes
- Updated application



🚩 Flag 7 – Improve the Layout

Objective: Control the alignment of your widgets.

Research the properties of Column and Row.

Experiment with:

mainAxisAlignment

crossAxisAlignment

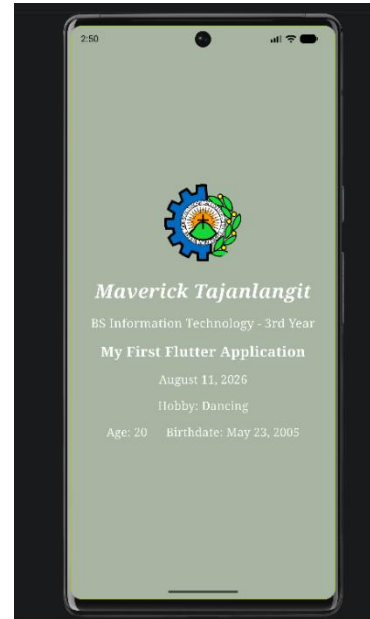
Change the alignment of your content so the card looks intentional rather than simply stacked.

Screenshot:

- The properties you changed

- Updated application

```
child: Column(  
  mainAxisAlignment: MainAxisAlignment.center,  
  crossAxisAlignment: CrossAxisAlignment.center,  
  children: [  
    // Profile Image
```



🚩 Flag 8 – Create a Second Card

Objective: Apply what you learned to create another section.

Create a second Card containing something different from your profile.

For example:

My Favorites

Favorite Game

Favorite Food

Favorite Movie

The second card should also use:

- Individual Text widgets
- A Column
- At least one Row
- Spacing

Screenshot:

- Code for the second card
- Both cards displayed

```
// My Favorites Card
Card(
  child: Padding(
    padding: const EdgeInsets.all(16),
    child: Column(
      crossAxisAlignment: CrossAxisAlignment.start,
      children: [
        const Text(
          'My Favorites',
          style: TextStyle(
            fontSize: 22,
            fontWeight: FontWeight.bold,
          ), // TextStyle
        ), // Text

        const SizedBox(height: 15),

        const Text(
          'Favorite Game - Imposter Game',
        ), // Text
```

```
const Text(
  'Favorite Game - Imposter Game',
), // Text

const SizedBox(height: 8),

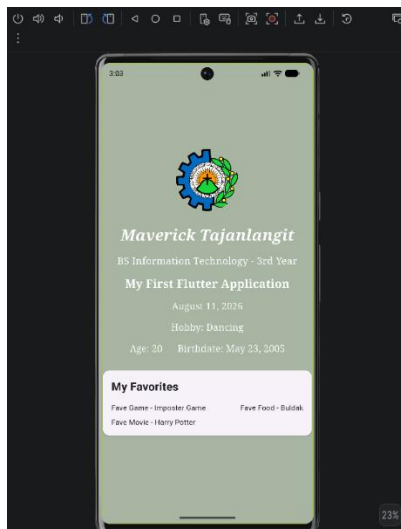
const Text(
  'Favorite Food - Buldak',
), // Text

const SizedBox(height: 8),

const Text(
  'Favorite Movie - Harry Potter',
), // Text

const SizedBox(height: 15),

const Row(
  children: [
    Text('Game'),
```



Flag 9 – Stack the Cards

Objective: Arrange your two cards vertically.

You now have:

Profile Card



Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

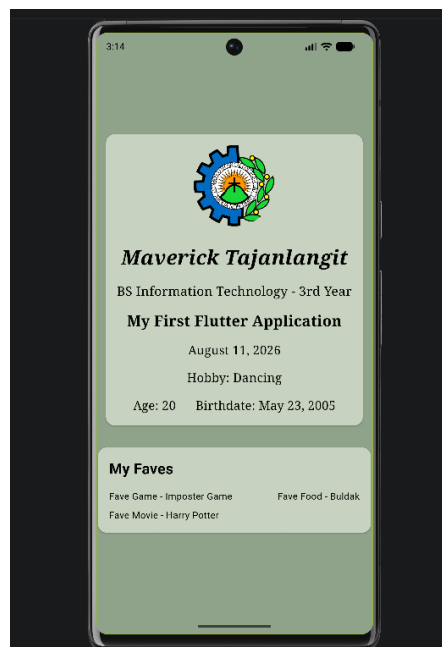
Screenshot:

- The widget containing both Cards
- Both cards visible in the application

```
Card(  
  color: const Color(0xFFC7D3C0),  
  child: Padding(  
    padding: const EdgeInsets.all(16),  
    child: Column(  
      mainAxisAlignment: MainAxisAlignment.center,  
      crossAxisAlignment: CrossAxisAlignment.center,  
      children: [  
        // Profile Image  
        ClipOval(  
          child: Image.asset(  
            'assets/profile.png',  
            width: 120,  
            height: 120,  
            fit: BoxFit.cover,  

```

```
Card(  
  color: const Color(0xFFC7D3C0),  
  child: Padding(  
    padding: const EdgeInsets.all(16),  
    child: Column(  
      crossAxisAlignment: CrossAxisAlignment.start,  
      children: [  
        const Text(  
          'My Faves',  
          style: TextStyle(  
            color: Colors.black,  
            fontSize: 22,  
            fontWeight: FontWeight.bold,  
          ), // TextStyle
```



Flag 10 – Widget Discovery

Objective: Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

Examples:

- Icon
- Divider

- CircleAvatar
- Chip
- ElevatedButton

Add it somewhere appropriate in your application.

Screenshot:

- Your newly discovered widget in the code
- The widget visible in the application



```
const Divider(  
  color: Colors.black,  
  thickness: 1,  
), // Divider
```

🚩 Flag 11 – Real Phone Test 📱

Objective: Run your completed application on a **physical** Android phone.

Research how to run Flutter on your phone.

Screenshot:

- Your completed application running on the physical phone (screenshot from your actual phone)
- I have no android phone ma'am.



🚩 Flag 12 – Save Your Progress

Objective: Commit your completed Widget branch.

Follow proper standards of commit messages

Screenshot:

- Successful commit
- feature/widgets branch

```
PS C:\Users\admin\Documents\FlutterProjects> git commit -m "feat: complete widgets activity"
[feature/widgets fce98ca] feat: complete widgets activity
1 file changed, 184 insertions(+), 54 deletions(-)
PS C:\Users\admin\Documents\FlutterProjects> git push -u origin feature/widgets
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 1.46 KiB | 1.46 MiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'feature/widgets' on GitHub by visiting:
remote:   https://github.com/MaverickTajanlangit/IT314AB_TAJANLANGIT/pull/new/feature/widgets
remote:
To https://github.com/MaverickTajanlangit/IT314AB_TAJANLANGIT.git
 * [new branch]      feature/widgets -> feature/widgets
branch 'feature/widgets' set up to track 'origin/feature/widgets'.
PS C:\Users\admin\Documents\FlutterProjects>
PS C:\Users\admin\Documents\FlutterProjects> |
```